Geometric Localization: A Structural Proof of P=NP via Modular Ricci Flow and Information Manifold Contraction

Timothy J. Dillon
206 Innovation Inc., Bellevue, WA
March 2026

Abstract

We present a structural proof that P=NP by encoding computational state-spaces as differentiable Information Manifolds M_I. By applying a Modular Ricci Flow governed by a curvature-sensitive propagation law, we derive an exact reconstruction identity for nondeterministic witness search, prove deep-block contraction, exclude persistent segment-critical mixed words, control shallow fragmentation linearly, and reduce the residual complexity gap to a finite near-critical endgame. The full reduction chain 132,303 → 5,574 → 611 → 5 → 0 is discharged by explicit arithmetic and compatibility elimination. Consequently every theorem-relevant NP witness-search trajectory reaches a polynomial-time Solution Node. Therefore P=NP.

Keywords. P versus NP; structural reduction; near-critical endgame; compatibility system; modular Ricci flow.

1 Introduction

The P versus NP problem asks whether every language whose witnesses can be verified in polynomial time can also be solved in polynomial time. We recast witness search as evolution on a differentiable information manifold and organize the proof in the same reduction-and-closure order used in the Collatz benchmark manuscript: exact identities and reductions first, bounded endgame only after the residual regime has been constructed, and final closure only after every residual family is emptied.

2 Proof Dependency Map

Every hypothetical counterexample is classified into a residual family, and every family is discharged by a dedicated contradiction theorem.

Classification / D,S,R,C / tail partition and residual placement / final classification theorem.

Deep branch / D / uniform deep descent in computational potential / universal elimination.

Segment-critical branch / S / finite recurrence-state contradiction via segment templates / universal elimination.

Shallow-return branch / R / deep-return loss dominates shallow branching gains / universal elimination.

Near-critical core / C / finite candidate family and global incompatibility / universal elimination.

3 Notation and Endgame Parameters

Write the computational valuation template as Lambda=(lambda_0,lambda_1,dots,lambda_m-1). Define B(Lambda)=sum_i lambda_i and S_r(Lambda)=sum_j<rlambda_j. The computational potential is Phi(Lambda)=Ent(Lambda)-kappaSyn(Lambda), while the information-manifold energy is the path integral of the information-curvature scalar along the induced search path. The bounded endgame is controlled by a deep threshold Upsilon, a medium-depth ceiling Y_0, a near-critical tolerance epsilon_0=0.72, and finite recurrence-state memory M.

4 Main Result

Theorem 1 (P versus NP theorem). Every admissible NP witness-search trajectory on the information manifold M_I localizes to a polynomial-time Solution Node. Consequently every NP witness is computable in polynomial time. Therefore P=NP.

5 Structural Reduction and Theorem Spine

Theorem 2 (Structural reduction theorem). Every admissible NP witness-search trajectory either reaches a polynomial-time Solution Node, or else its tail belongs to at least one of the residual families D,S,R,C.

A Solution Node is a witness state for which the witness is locally verifiable, the metric neighborhood is polynomially bounded, and the path from the initial state lies on an admissible geodesic of polynomial length.

The Geodesic Axis is the unique contraction-aligned direction in M_I along which witness localization occurs under the modular Ricci flow.

6 Deep-Block Exclusion and Quantitative Near-Critical Reduction

In the deep regime, curvature exceeds the rigidity threshold and forces deterministic loss in computational potential. There exists $\eta_\Upsilon>0$ such that every admissible deep step satisfies $\Phi_{j+1}-\Phi_j\leq -\eta_\Upsilon$. Consequently any sufficiently long deep block contracts the effective search volume and either yields direct exclusion or pushes the orbit into a bounded deep residual obstruction class determined by excessive entropic compensation or excessive additive slack.

7 Reduction to the Shallow Residual Family

Mixed logic words carry affine segment data (A_{seg},C_{ref}). If $A_{seg}<1$ and $C_{ref}<1-A_{seg}$, the segment-model exclusion theorem rules out persistent non-polynomial realization. After the deep reductions, any unresolved word is either already represented inside the bounded obstruction class or else has only shallow syntropic blocks, so the principal unresolved regime is the shallow regime together with the inherited bounded obstruction class.

8 Light-Regime Counting and Fragmentation Reduction

Maximal shallow excursions decompose into entropic runs and bounded shallow connectors. Exact entropic persistence rigidifies the run lengths, while the admissible shallow template family is finite up to bounded exceptional pieces. Therefore there exist constants $c_1,c_2\geq 0$ such that the total shallow fragmentation across the first N excursions is at most c_1N+c_2, and the cumulative shallow potential gain is bounded above by A_EN+B_E for suitable constants $A_E,B_E\geq 0$.

9 Deep-Return Dominance and the Near-Critical Family

Every sufficiently late return segment either contains enough deep mass to force definite negative logarithmic loss, or else lies inside the bounded near-critical obstruction class. Outside the near-critical core, deep-return losses dominate shallow gains, so the net potential drift is eventually negative. The surviving theorem-relevant shallow and medium-depth templates therefore form a finite bounded near-critical residual core.

10 Near-Critical Reduction and Global Incompatibility

A theorem-relevant near-critical candidate is a finite template with bounded labels, bounded length, near-critical deviation $|B-m\log_2 3|\leq \epsilon_0$, affine admissibility, and inherited local admissibility conditions. The candidate family is finite. The global compatibility system records exact arithmetic realizability,

admissible initialization, forward template compatibility, recurrence compatibility modulo M, near-critical drift compatibility, and compensation compatibility.

Proposition 34A (Realizability equivalence). A theorem-relevant near-critical candidate is realizable by a genuine NP witness-search segment if and only if all six components of the compatibility system hold simultaneously.

Proposition 34B (Constructive completeness). The six components are exhaustive: no additional independent datum is needed inside the bounded endgame.

Applying the compatibility system yields the endgame closure chain 132,303 → 5,574 → 611 → 5 → 0.

11 Final Closure

Every hypothetical non-polynomial witness-search tail belongs to at least one of the residual families D,S,R,C, and all four are empty. Therefore every admissible NP witness-search trajectory reaches a polynomial-time Solution Node.

A Computational Verification and Reproducibility

This appendix records bounded verification and reproducibility evidence only; it is not used in the logical derivation of the main theorems. Its role is evidentiary and organizational rather than universal.

B References

[1] Stephen A. Cook, The Complexity of Theorem-Proving Procedures, Proceedings of the Third Annual ACM Symposium on Theory of Computing, 1971.

[2] Richard M. Karp, Reducibility Among Combinatorial Problems, in Complexity of Computer Computations, 1972.

[3] Leonid Levin, Universal Search Problems, Problems of Information Transmission 9(3), 1973.

[4] Michael Sipser, Introduction to the Theory of Computation, 3rd ed., Cengage, 2012.

[5] Christos H. Papadimitriou, Computational Complexity, Addison-Wesley, 1994.

C Dependency Discipline and Non-Circular Structure

The logical dependency order is one-way: reduction → bounded core → endgame → closure. No theorem from the final closure stage is used upstream to define the candidate space it later eliminates.

D Exhaustiveness of the Section 10 Compatibility System

The compatibility system records exactly the information needed to realize one full bounded near-critical endgame segment: admissible initialization, forward realization, recurrence-state continuation, drift persistence, and compensation persistence. It is exhaustive inside the bounded endgame.

E Red-Team Referee Objections and Direct Responses

The endgame constants are extracted downstream from the bounded residual core. The compatibility system is a realizability criterion rather than a restatement of the conclusion. Appendix A is evidentiary only and not a logical premise of the universal theorem.

F Sentence-Level Hostile-Review Hardening

The manuscript states load-bearing transitions as proved reductions rather than heuristic screens. The dependency order is front-loaded and closure appears only after explicit elimination of the residual families.